Lifting an odd singleton budget across the opposite end

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$ be a transitive connected string C-group, $n = 2r \ge 14$, with every maximal omission intransitive. Suppose, as actual permutation subgroups, that

$$G_0 = \mathrm{Alt}(\Omega \setminus \{a\}), \qquad C = G_{r-1} = \mathrm{Alt}(\Omega \setminus \{d\}). \tag{1}$$

Assume every colour $2, \dots, r-2$ has at least two ambient-eligible edges. An ambient-eligible j-edge joins distinct G_j-orbits. All the previously accepted ambient interval and selected-localization proofs, including their complete retained inputs, are required.

Theorem 1. *Under these hypotheses either $G_{>1}$ fixes at least three points or $G_{>2}$ fixes at least five. Equivalently there is $h \in \{1, 2\}$ such that, for $X_h = \mathrm{supp}(G_{>h})$,*

$$2h \le n - |X_h| - 1. \tag{2}$$

The fixed points in this conclusion belong to the full ambient tail, not merely the corresponding subgroup of C.

Proof. Put $\eta = \rho_{r-1}$, $\sigma = \rho_1$, $b = a^{\rho_0}$,

$$H_C = G_{\{2,\dots,r-2\}}, \quad K_C = G_{\{3,\dots,r-2\}}, \quad H = \langle H_C, \eta \rangle, \quad K = \langle K_C, \eta \rangle.$$

The points a, d are distinct, since otherwise G fixes their common point. The generator η moves d into the natural C-orbit, and fixes a. Because it commutes with ρ_0, it also fixes b. Similarly ρ_0 fixes d. Thus $\rho_0 = (a\ b)\delta$, where δ is an odd involution fixing a, b, d, and it centralizes H_C and η. The unique ambient-eligible η-edge is the one incident with d.

We first justify the use of the accepted odd singleton theorem in C. Its degree is $n - 1 = 2(r-1) + 1$. Its marking is connected: otherwise IP decomposes this nonabelian simple group as a direct product of nontrivial marked factors. For any omitted colour $j \le r-2$, if C_j were transitive on $\Omega \setminus \{d\}$, then $\langle C_j, \eta \rangle \le G_j$ would be transitive on Ω, which is excluded. Its first omission is the actual natural alternating group on $\Omega \setminus \{a, d\}$, by IP and (1). Every specified ambient-eligible edge lies in the C-domain and is also C_j-eligible, since $C_j \le G_j$. This is only the valid global-to-local direction.

Let f_C be the number of H_C-fixed points in the C-domain, and let c be its nontrivial orbit count. The local interval bound and the accepted odd theorem give

$$f_C + c \le 5, \qquad f_C \ge 3 \quad \text{or} \quad \begin{cases} f_C = 2,\ H_C \text{ has orbits } \{a\}, \{b\}, E, V, \\ E = \{u, v\},\ |V| = n - 5 \text{ odd},\ \rho_0 = (a\ b)(u\ v). \end{cases} \tag{3}$$

Here $c \ge 1$, since the marked subgroup H_C is nontrivial. We now analyze the extra fixed points rather than assume they remain fixed on adjoining η.

Suppose $f_C = 3$, with additional fixed point z. The involution δ fixes z. If it is free on the entire nontrivial H_C-support (including the possibility of exchanging two orbits), its full fixed set is $\{a, b, z, d\}$. Since η centralizes δ, fixes a, b and moves d, it swaps d, z. But σ fixes d and centralizes η, so fixes z too. Consequently $G_0 = \langle \sigma, H_C, \eta \rangle$ preserves $\{d, z\}$, contrary to (1).

The only other possibility has two preserved nontrivial orbits: O of size $2q$, on which δ is free, and V of even size v, on which it is the identity. Oddness makes q odd. Partition the colours

$2, \dots, r-1$ into L, those with an ambient-eligible edge in O, and F, the rest. The projected δ-pair forest gives $|L| \le q-1$: any alternate path of pair-blocks lifts using $\rho_0 \in G_j$ and violates the selected ambient omission cut. The colour $r-1$ belongs to F, since its only eligible edge is incident with the δ-fixed point d. Every eligible edge of $F \setminus \{r-1\}$ is confined to $W = V \cup \{z, d\}$, which has three H_C-orbits, including both singletons. Selected localization in the full interval $\{2, \dots, r-2\}$ gives

$$2(|F|-1) \le |W| - o_{H_C}(W) = v-1, \qquad |F| \le v/2.$$

Thus $r-2 = |L|+|F| \le q-1+v/2$, contradicting $n = 4+2q+v = 2r$. This excludes $f_C = 3$.

If $f_C = 4$, then $c = 1$ and its nontrivial orbit has odd size. The centralizing involution δ fixes that orbit pointwise. Its oddness therefore forces it to swap the other two fixed points, say $E = \{u, v\}$. Again $\rho_0 = (a\ b)(u\ v)$, but now H_C fixes E pointwise. If η also fixes E, then H has at least four fixed points and (2) holds with $h = 1$. Values $f_C \ge 5$ are impossible by (3) and $c \ge 1$.

It remains to consider the literal double-transposition head from either surviving case. In both cases H_C preserves E and is transitive on the odd remaining set V. In the $f_C = 2$ case, the second proof of the accepted odd theorem proves explicitly that K_C fixes E pointwise; if it did not, the two pair images and the conjugate ρ_0^σ would confine all its eligible edges to a set of size exactly twice its rank, contradicting strict localized counting. In the $f_C = 4$ case this fixation is immediate. Hence in both cases K_C fixes

$$a, b, E, E^\sigma,\ x = b^\sigma. \tag{4}$$

Transitivity of $C_0 = \langle \sigma, H_C \rangle$ gives $x \ne b$; σ fixes a, d, so $x \in E \cup V$. Also η fixes x, since it centralizes σ and fixes b.

Suppose η fixes E pointwise. It then fixes E^σ. If $x \in V$, (4) gives five distinct K-fixed points a, b, u, v, x. If $x \in E$, rename it u; then $u^\sigma = b$, and $w = v^\sigma$ belongs to V. Indeed $w = v$ would make $\{b, u, v\}$ invariant under C_0, and the other excluded values follow from injectivity and fixation of a, d. Now a, b, u, v, w are five distinct K-fixed points. This gives (2) with $h = 2$.

Finally suppose η swaps E. Then $x \notin E$. The pair E^σ is also η-invariant. It cannot equal E, as that would make E invariant under C_0. Their intersection cannot be a singleton, because η would fix that singleton and hence both points of E. Thus they are disjoint, and $E^\sigma = \{w, z\} \subset V \setminus \{x\}$. By (4), K_C fixes the seven distinct points a, b, x, u, v, w, z in the C-domain, as well as d. All its actual edges, and hence all its ambient-eligible edges, are confined to the invariant complement Y of these eight points. Every one of its $r-4$ colours has two such eligible edges. But $|Y| = n-8 = 2(r-4)$, and the localized interval theorem gives

$$2(r-4) \le |Y| - o_{K_C}(Y) < |Y|,$$

a contradiction. This completes all cases and proves the ambient fixed-support budget. □

Attribution and remaining scope. This supplies a precise two-singleton-end embedding step for Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.32 (physical pages 44–45). The source's fixed-support budget strategy and the complete accepted odd singleton and selected-localization proofs are reused. The additional argument proves which local fixed points survive the opposite generator; it makes no general parabolic inheritance claim. The two natural alternating omissions, the even critical degree and the paired interior colours are explicit hypotheses. Other opposite heads, final budget combination, source equality diagrams and the remaining rigidity/table contracts are not recovered here. No new computation, formal statement, original or question credit is claimed.